Motion analysis

Horizontal toss

Source	Synthetic projectile demo
Object	Not specified
Mass	0.2 kg (supplied by the operator)
Samples	66 measured; 66 modeled
Duration	0.650 s
Maximum y	3.051 m (chosen datum)
Maximum speed	5.555 m/s (modeled)

Physics estimates

Gravity: 9.81 m/s²
Gravity fit standard error: 0.00244 m/s². This excludes calibration and other measurement systematics.
Restitution: not established by qualifying measurements.

Method

Ballistic intervals use least-squares quadratic position models in each axis against the recorded timestamps. Velocity and acceleration are analytic derivatives.
Missing observations and interval boundaries are not interpolated. Potential energy uses g = 9.81 m/s² and the chosen y = 0 datum; kinetic energy uses modeled speed. Fit standard errors, when present in the CSV/JSON, do not include calibration or camera systematics.

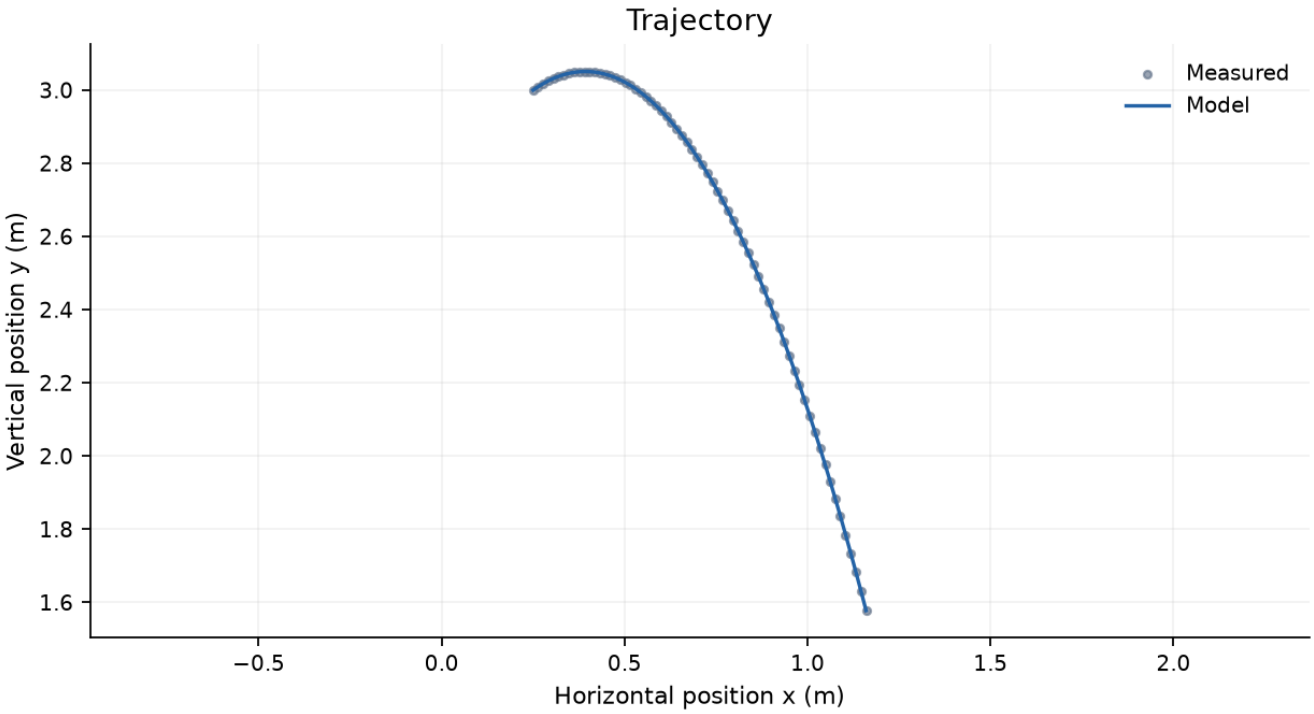
Measurement notes

- Positions assume planar motion and a fixed, level camera; perspective, depth, and lens distortion are uncorrected.
- Potential energy uses measured Y and the chosen y=0 datum, not a measured ground level; energies use g=9.81 m/s².
- Fit standard errors assume independent, equal-variance position residuals, exact timestamps, and the chosen model; they are not measurement uncertainty and exclude calibration, tracking, and timing errors.
- Ballistic estimates assume the selected interval is unforced free flight. A high R² cannot establish that physical assumption.

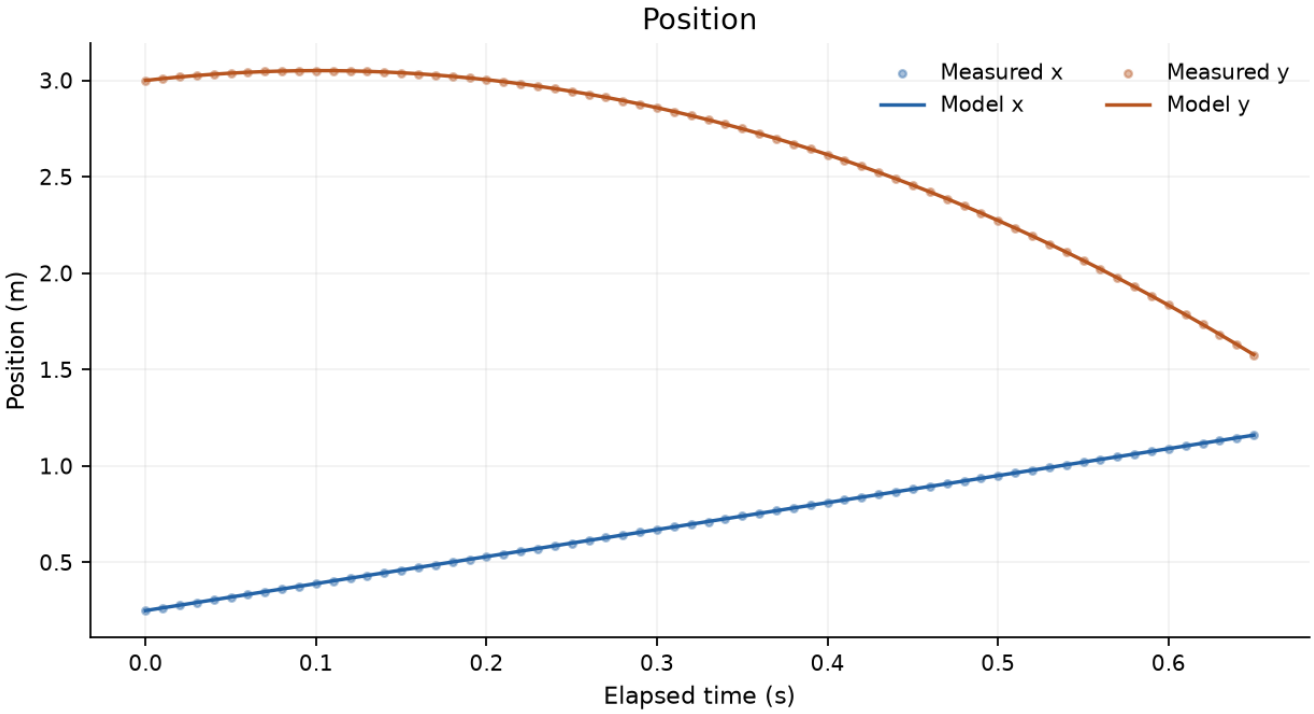
Fitting intervals

Interval	Time (s)	Samples	R² x	R² y
1	0.000–0.650	66	1.0000	1.0000

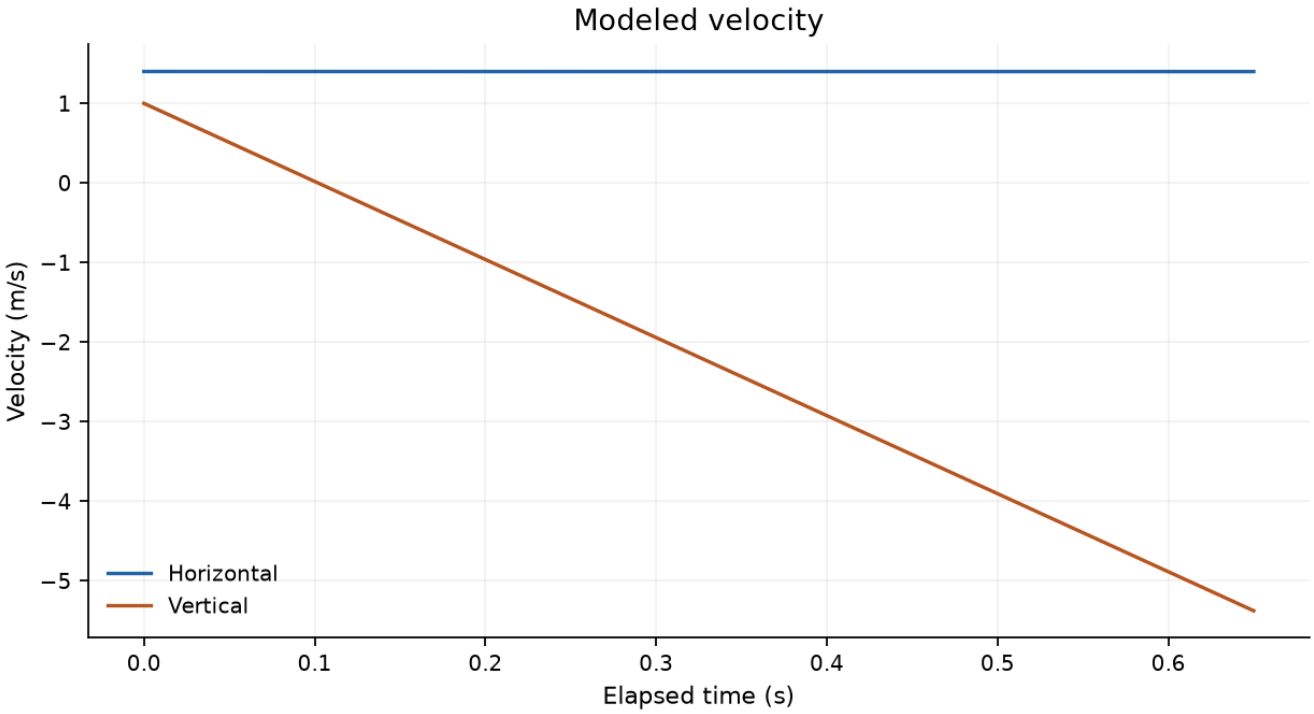
Trajectory



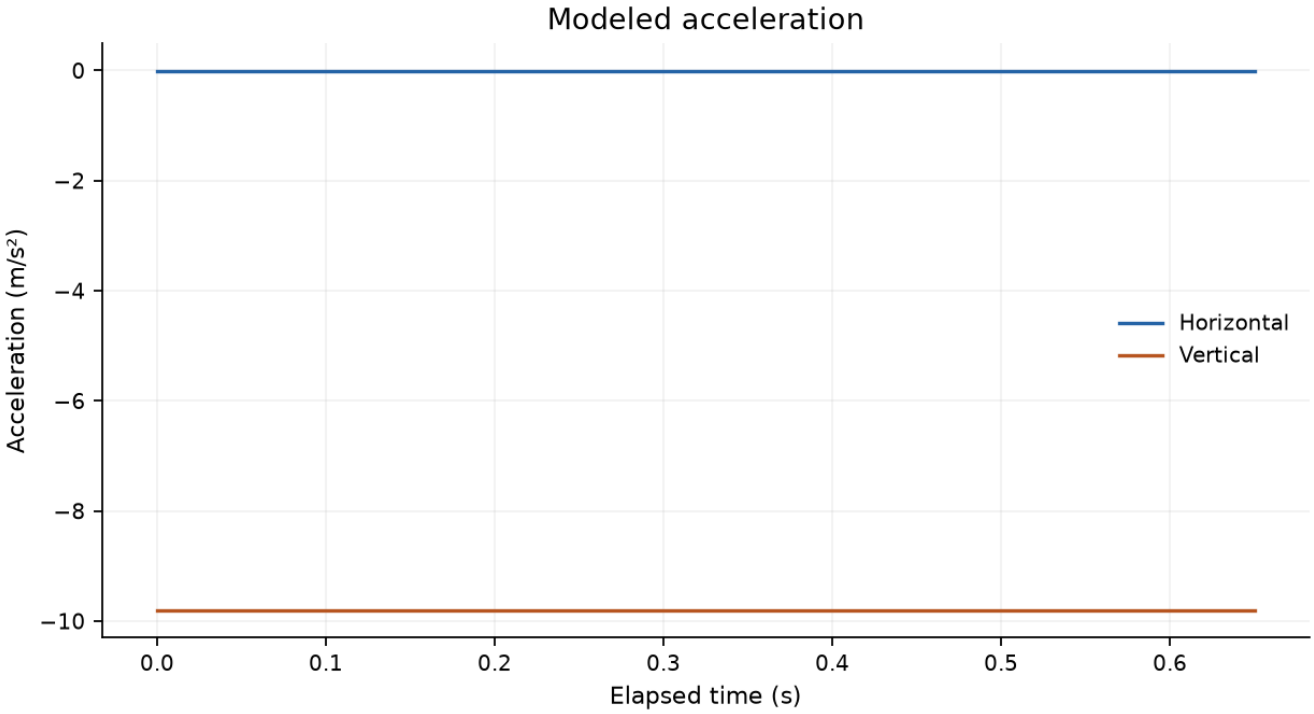
Position over time



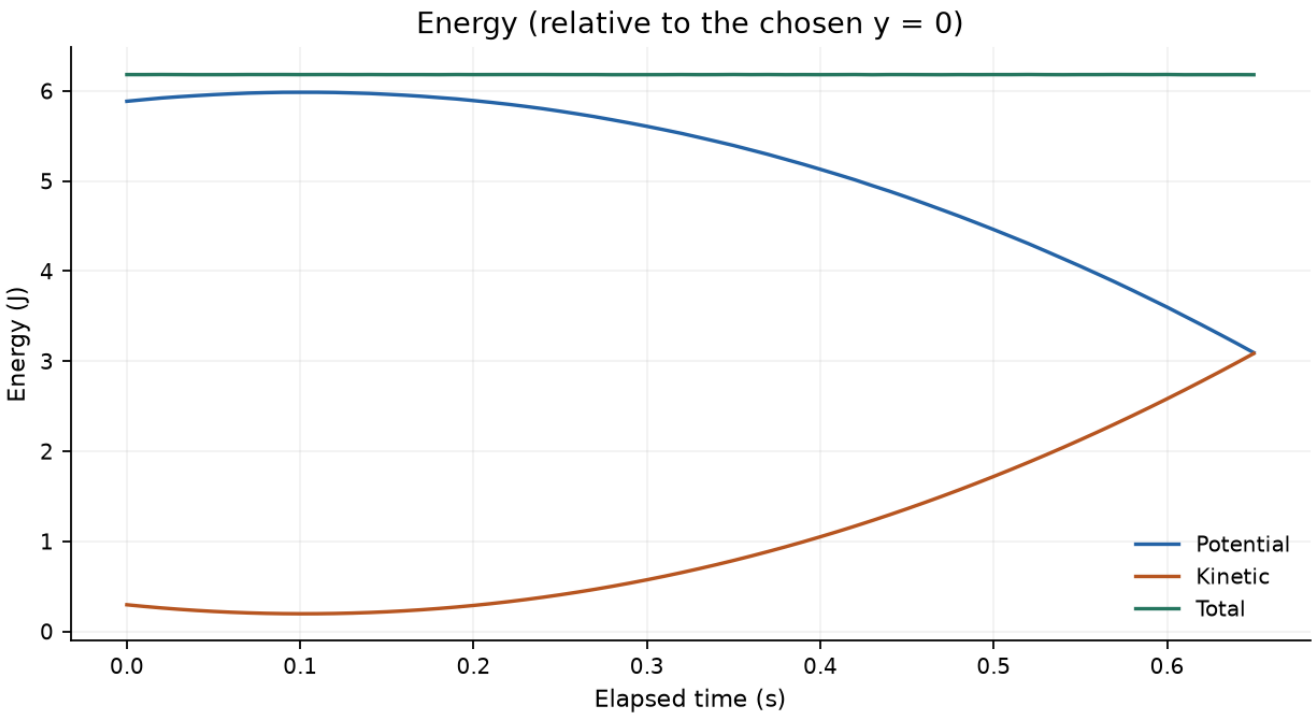
Velocity



Acceleration



Energy



Fitting intervals

